

KANGWA MUKUKA

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A Student of Computer Science, passionate about research and driving social impact in resource-constrained settings. As a research assistant at DataLab Research Group, the team is reimaging what we think is possible with technology in Zambia and working on projects at the cusp of Machine Learning and Artificial Intelligence for Healthcare in the country.

EDUCATION

Bachelor of Science - BSc., Computer Science | Major: Computer Systems Engineering Feb 2022 - May 2026
University of Zambia (UNZA) | Lusaka, ZM

Prj: *Enhancing Radiology Workflows: Semi-automated Cervical Cancer Reporting at the Cancer Diseases Hospital (Zambia)*

Supervisors: [Dr. Lighton Phiri](#), [Dr. Mayumbo Nyirenda](#)

HONORS AND AWARDS

Undergraduate Student Innovator of the Year Award UNZA (2025), [African Network for Artificial Intelligence in Biomedical Imaging \(AFRICAI\) Mentorship Award \(2025\)](#), 3rd place: IEEE IAS IPCSD Engineering Design Contest Africa (2024), 3rd place: Zambia@60 National ICT Hackathon (2024), 1st place: UNZA Student Innovation Challenge (2025).

RELEVANT COURSEWORK

Fundamentals of Artificial Intelligence | Data Mining & Warehousing | IT Project Management | Computer Programming

SKILLS

Technical Skills: Python, JavaScript, Java, C++, Flask, React, CSS, Bootstrap, MySQL, PyTorch.

Tools: Git, GitHub, Firebase, Google Colab, Jupyter Notebook.

Soft Skills: Research Methods, Requirements gathering, Teamwork, Collaboration, Troubleshooting, Public speaking.

WORK AND LEADERSHIP EXPERIENCE

Research Assistant - Enterprise Medical Imaging | DataLab Research Group Sep 2025 - Present
University of Zambia | Lusaka, ZM

- Researching the use of Vision Language Models to semi-automate cervical cancer report generation at the Cancer Diseases Hospital with the aim of reducing the time to 1/3, addressing the 3,600+ annual cervical cancer cases in Zambia, with the use of AI and Machine Learning. Selected as one of 10 mentees across Africa by [AFRICAI](#).
- Secured key collaboration with [Dawa Health](#), working towards a unified screening tool for the early detection of cervical cancer among women in Zambia, Zimbabwe, and Rwanda, where 8,000+ combined cases occur annually.

Chairperson

Dec 2024 - Nov 2025

Institute of Electrical and Electronics Engineers (IEEE) UNZA Student Branch | Lusaka, ZM

- Led an executive team of 11 students and an association of 560+ student members. Our team won the first continental contest for the Student Branch, emerging 3rd in Africa. Facilitated industrial tours promoting STEM.

Technical Lead

Sep 2023 - Jun 2024

Google Developer Student Club (GDSC) UNZA Student Chapter | Lusaka, ZM

- Led 55+ students in technical activities by conducting information sessions and hosting fireside chats with inspiring Software Engineering industry leaders. I actively recruited GDSC members during 1st year orientation.

FEATURED PROJECTS

Mota50 - Fleet Management System | World Vision Zambia (WVZ) Innovation Challenge Nov 2025 - Present

- Built to streamline fleet management operations at WVZ, innovating for development challenges in Zambia.

Enterprise Medical Imaging in Zambia (EMI x Z) | AI Supported Workflows [\[emi.org.zm\]](#) Mar 2025 - Present

- This project seeks to investigate the challenges resulting from the shortage of radiologists and to investigate the feasibility of implementing efficient and effective medical imaging workflows using enterprise imaging techniques.

ReviZa | Transforming the Future of Education in Zambia [\[reviza-31466.web.app\]](#) Aug 2024 - Mar 2025

- Spearheaded product/strategy of this app to facilitate access to past papers & academic resources. Gained 100+ active users in the 1st month. Emerged 3rd of 89 teams in the National Hackathon by the [Ministry of Education](#).

PUBLICATIONS | PEER-REVIEWED CONFERENCE PAPERS

Design and Prototyping of a Remote Livestock Monitoring Device for Low-Resource Settings

- M. M. Makumba, J. C. Moya, D. Banda and K. E. Mukuka, IEEE International Humanitarian Technology Conference (IHTC), Edmonton, AB, Canada, 2025, pp. 44-48, doi: [10.1109/IHTC65087.2025.11216306](https://doi.org/10.1109/IHTC65087.2025.11216306).

Design of a Low-Cost Smart Solar-Powered Irrigation System

- K. M. Makasa and K. E. Mukuka, IEEE PES/IAS PowerAfrica Conference (PAC), Johannesburg, South Africa, 2024, pp. 1-5, doi: [10.1109/PowerAfrica61624.2024.10759498](https://doi.org/10.1109/PowerAfrica61624.2024.10759498).